ROCSCIENCE

Case11
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case11

Project Summary

File Name:	Case11.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.561s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.005
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

soil 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	0 kPa
Phi	38 °
Water Surface	None
Ru Value	0

soil 2	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	5.3 kPa
Phi	23 °
Water Surface	None
Ru Value	0

soil 3	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	7.2 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.404400
Center:	34.356, 43.178
Radius:	18.708
Left Slip Surface Endpoint:	29.932, 25.000
Right Slip Surface Endpoint:	51.183, 35.000
Resisting Moment:	14071.7 kN-m
Driving Moment:	10019.7 kN-m
Total Slice Area:	72.0233 m2
Surface Horizontal Width:	21.2507 m
Surface Average Height:	3.38922 m

Method: spencer

FS	1.374150
Center:	34.248, 43.117
Radius:	18.621
Left Slip Surface Endpoint:	29.943, 25.000
Right Slip Surface Endpoint:	51.007, 35.000
Resisting Moment:	13381.7 kN-m
Driving Moment:	9738.19 kN-m
Resisting Horizontal Force:	623.119 kN
Driving Horizontal Force:	453.459 kN
Total Slice Area:	70.0235 m2
Surface Horizontal Width:	21.0641 m
Surface Average Height:	3.32431 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	11113
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	10914
Number of Invalid Surfaces:	199

Error Codes

Error Code -108 reported for 119 surfaces
Error Code -111 reported for 75 surfaces
Error Code -112 reported for 5 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.4044

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.435262	1.08444	-12.9953	soil 3	7.2	20	6.13853	8.62095	3.90402	0	3.90402	2.48736	2.48736
2	0.435262	3.71568	-11.6308	soil 3	7.2	20	7.75143	10.8861	10.1275	0	10.1275	8.53202	8.53202
3	0.435262	6.27784	-10.2729	soil 3	7.2	20	9.30028	13.0613	16.1038	0	16.1038	14.4182	14.4182
4	0.435262	8.74974	-8.92083	soil 3	7.2	20	10.7735	15.1303	21.7884	0	21.7884	20.0973	20.0973
5	0.435262	11.1325	-7.57374	soil 3	7.2	20	12.1735	17.0965	27.1904	0	27.1904	25.5717	25.5717
6	0.435262	13.4269	-6.23086	soil 3	7.2	20	13.5023	18.9627	32.3178	0	32.3178	30.8436	30.8436
7	0.435262	15.6338	-4.89141	soil 3	7.2	20	14.7619	20.7316	37.1779	0	37.1779	35.9146	35.9146
8	0.435262	17.7538	-3.55463	soil 3	7.2	20	15.9539	22.4056	41.777	0	41.777	40.7859	40.7859
9	0.435262	19.7873	-2.21979	soil 3	7.2	20	17.0796	23.9866	46.1207	0	46.1207	45.4587	45.4587
10	0.435262	21.7347	-0.886154	soil 3	7.2	20	18.1406	25.4766	50.2145	0	50.2145	49.9339	49.9339
11	0.435262	23.596	0.447002	soil 3	7.2	20	19.1377	26.877	54.0621	0	54.0621	54.2114	54.2114
12	0.435262	25.3714	1.7804	soil 3	7.2	20	20.0721	28.1893	57.6677	0	57.6677	58.2916	58.2916
13	0.435262	27.0606	3.11476	soil 3	7.2	20	20.9447	29.4147	61.0342	0	61.0342	62.1739	62.1739
14	0.435262	28.6635	4.45082	soil 3	7.2	20	21.756	30.5541	64.1648	0	64.1648	65.8582	65.8582
15	0.435262	30.1796	5.78931	soil 3	7.2	20	22.5066	31.6083	67.0613	0	67.0613	69.3432	69.3432
16	0.435262	31.6084	7.13098	soil 3	7.2	20	23.1971	32.578	69.7253	0	69.7253	72.6274	72.6274
17	0.435262	32.9492	8.47659	soil 3	7.2	20	23.8277	33.4636	72.1586	0	72.1586	75.7097	75.7097
18	0.435262	34.2011	9.82694	soil 3	7.2	20	24.3987	34.2655	74.3619	0	74.3619	78.5881	78.5881
19	0.435262	35.3631	11.1828	soil 3	7.2	20	24.9101	34.9837	76.3352	0	76.3352	81.2598	81.2598
20	0.435262	36.4341	12.5451	soil 3	7.2	20	25.362	35.6184	78.0788	0	78.0788	83.7224	83.7224
21	0.435262	37.4126	13.9146	soil 3	7.2	20	25.7542	36.1692	79.5921	0	79.5921	85.9726	85.9726
22	0.435262	38.2971	15.2923	soil 3	7.2	20	26.0864	36.6358	80.8741	0	80.8741	88.0068	88.0068
23	0.435262	39.0857	16.6791	soil 3	7.2	20	26.3584	37.0177	81.9236	0	81.9236	89.821	89.821
24	0.435262	39.7766	18.076	soil 3	7.2	20	26.5696	37.3143	82.7384	0	82.7384	91.4104	91.4104
25	0.435262	40.3673	19.4841	soil 3	7.2	20	26.7194	37.5247	83.3162	0	83.3162	92.7697	92.7697
26	0.435262	40.8554	20.9047	soil 3	7.2	20	26.807	37.6477	83.6545	0	83.6545	93.8935	93.8935
27	0.435262	41.238	22.3388	soil 3	7.2	20	26.8315	37.6822	83.7491	0	83.7491	94.7748	94.7748
28	0.435262	41.5119	23.7878	soil 3	7.2	20	26.7919	37.6266	83.5965	0	83.5965	95.4064	95.4064
29	0.435262	41.6735	25.2532	soil 3	7.2	20	26.6871	37.4793	83.1916	0	83.1916	95.7799	95.7799
30	0.407378	39.0484	26.6883	soil 2	5.3	23	28.4369	39.9368	81.5991	0	81.5991	95.8942	95.8942
31	0.407378	38.9893	28.0937	soil 2	5.3	23	28.1695	39.5612	80.7143	0	80.7143	95.7514	95.7514
32	0.407378	38.8276	29.5177	soil 2	5.3	23	27.8322	39.0875	79.5983	0	79.5983	95.3564	95.3564
33	0.407378	38.5587	30.9621	soil 2	5.3	23	27.4232	38.5132	78.2455	0	78.2455	94.6984	94.6984
34	0.407378	38.178	32.4287	soil 2	5.3	23	26.9409	37.8358	76.6496	0	76.6496	93.7657	93.7657
35	0.407378	37.68	33.9195	soil 2	5.3	23	26.3829	37.0522	74.8034	0	74.8034	92.5451	92.5451
36	0.407378	37.0585	35.437	soil 2	5.3	23	25.7468	36.1588	72.6987	0	72.6987	91.021	91.021
37	0.412604	36.7664	36.9937	soil 1	0	38	34.9615	49.0999	62.845	0	62.845	89.1843	89.1843
38	0.412604	35.851	38.5932	soil 1	0	38	33.505	47.0544	60.2269	0	60.2269	86.9671	86.9671
39	0.412604	34.782	40.2292	soil 1	0	38	31.9188	44.8267	57.3755	0	57.3755	84.3768	84.3768
40	0.412604	33.5482	41.9057	soil 1	0	38	30.1994	42.412	54.2848	0	54.2848	81.3866	81.3866
41	0.412604	32.1362	43.6276	soil 1	0	38	28.3429	39.8048	50.9477	0	50.9477	77.9644	77.9644
42	0.412604	30.5307	45.4004	soil 1	0	38	26.345	36.9989	47.3564	0	47.3564	74.0722	74.0722
43	0.412604	28.7129	47.2307	soil 1	0	38	24.2009	33.9878	43.5024	0	43.5024	69.6652	69.6652
44	0.412604	26.6603	49.1268	soil 1	0	38	21.9053	30.7638	39.376	0	39.376	64.688	64.688
45	0.412604	24.3452	51.0985	soil 1	0	38	19.4526	27.3192	34.9669	0	34.9669	59.0734	59.0734
46	0.412604	21.7327	53.1584	soil 1	0	38	16.8367	23.6455	30.2648	0	30.2648	52.7369	52.7369
47	0.412604	18.7779	55.3227	soil 1	0	38	14.0518	19.7344	25.2589	0	25.2589	45.5695	45.5695
48	0.412604	14.7981	57.613	soil 1	0	38	10.644	14.9484	19.133	0	19.133	35.9136	35.9136
49	0.412604	9.31425	60.0586	soil 1	0	38	6.39738	8.98448	11.4996	0	11.4996	22.6064	22.6064
50	0.412604	3.21624	62.7024	soil 1	0	38	2.09002	2.93523	3.75693	0	3.75693	7.80669	7.80669

Global Minimum Query (spencer) - Safety Factor: 1.37415

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.416371	1.0111	-12.7103	soil 3	7.2	20	7.22923	9.93404	7.51171	0	7.51171	5.88118	5.88118
2	0.416371	3.40769	-11.4001	soil 3	7.2	20	9.13888	12.5582	14.7215	0	14.7215	12.8788	12.8788
3	0.416371	5.7398	-10.096	soil 3	7.2	20	10.9243	15.0116	21.4622	0	21.4622	19.5171	19.5171
4	0.416371	7.99266	-8.7971	soil 3	7.2	20	12.5811	17.2883	27.7174	0	27.7174	25.7704	25.7704
5	0.416371	10.1672	-7.50278	soil 3	7.2	20	14.1173	19.3993	33.5174	0	33.5174	31.6581	31.6581
6	0.416371	12.2641	-6.2123	soil 3	7.2	20	15.5402	21.3545	38.8892	0	38.8892	37.1976	37.1976
7	0.416371	14.284	-4.92497	soil 3	7.2	20	16.856	23.1627	43.8571	0	43.8571	42.4046	42.4046
8	0.416371	16.2275	-3.64014	soil 3	7.2	20	18.0705	24.8316	48.4424	0	48.4424	47.2928	47.2928
9	0.416371	18.095	-2.35713	soil 3	7.2	20	19.1888	26.3683	52.6644	0	52.6644	51.8745	51.8745
10	0.416371	19.8866	-1.07531	soil 3	7.2	20	20.2155	27.7791	56.5408	0	56.5408	56.1613	56.1613
11	0.416371	21.6025	0.205969	soil 3	7.2	20	21.1548	29.0698	60.0868	0	60.0868	60.1629	60.1629
12	0.416371	23.2429	1.48735	soil 3	7.2	20	22.0103	30.2454	63.3168	0	63.3168	63.8883	63.8883
13	0.416371	24.8075	2.76948	soil 3	7.2	20	22.7855	31.3107	66.2436	0	66.2436	67.3459	67.3459
14	0.416371	26.2963	4.05301	soil 3	7.2	20	23.4835	32.2698	68.8787	0	68.8787	70.5426	70.5426
15	0.416371	27.7089	5.33857	soil 3	7.2	20	24.1068	33.1264	71.2323	0	71.2323	73.485	73.485
16	0.416371	29.0449	6.62683	soil 3	7.2	20	24.6582	33.8841	73.3141	0	73.3141	76.1788	76.1788
17	0.416371	30.3037	7.91847	soil 3	7.2	20	25.1398	34.5459	75.1323	0	75.1323	78.629	78.629
18	0.416371	31.4847	9.21418	soil 3	7.2	20	25.5537	35.1146	76.6947	0	76.6947	80.84	80.84
19	0.416371	32.5871	10.5147	soil 3	7.2	20	25.9017	35.5928	78.0084	0	78.0084	82.8159	82.8159
20	0.416371	33.6099	11.8206	soil 3	7.2	20	26.1854	35.9826	79.0795	0	79.0795	84.5597	84.5597
21	0.416371	34.5521	13.1329	soil 3	7.2	20	26.4063	36.2862	79.9137	0	79.9137	86.0746	86.0746
22	0.416371	35.4124	14.4522	soil 3	7.2	20	26.5657	36.5053	80.5156	0	80.5156	87.3623	87.3623
23	0.416371	36.1894	15.7794	soil 3	7.2	20	26.6649	36.6416	80.89	0	80.89	88.4251	88.4251
24	0.416371	36.8816	17.1153	soil 3	7.2	20	26.7049	36.6965	81.0409	0	81.0409	89.2642	89.2642
25	0.416371	37.4871	18.4609	soil 3	7.2	20	26.6864	36.6711	80.9713	0	80.9713	89.8802	89.8802
26	0.416371	38.004	19.8172	soil 3	7.2	20	26.6104	36.5667	80.6844	0	80.6844	90.2737	90.2737
27	0.416371	38.4301	21.1851	soil 3	7.2	20	26.4775	36.384	80.1823	0	80.1823	90.4443	90.4443
28	0.416371	38.7628	22.5658	soil 3	7.2	20	26.2881	36.1238	79.4674	0	79.4674	90.3917	90.3917
29	0.416371	38.9995	23.9605	soil 3	7.2	20	26.0427	35.7866	78.5411	0	78.5411	90.1145	90.1145
30	0.416371	39.1371	25.3704	soil 3	7.2	20	25.7417	35.373	77.4046	0	77.4046	89.6114	89.6114
31	0.455654	42.864	26.8652	soil 2	5.3	23	27.2727	37.4768	75.8038	0	75.8038	89.6192	89.6192
32	0.455654	42.7661	28.4484	soil 2	5.3	23	26.7254	36.7247	74.0318	0	74.0318	88.5114	88.5114
33	0.455654	42.5223	30.0556	soil 2	5.3	23	26.102	35.868	72.0136	0	72.0136	87.1174	87.1174
34	0.455654	42.1255	31.6894	soil 2	5.3	23	25.4023	34.9066	69.7488	0	69.7488	85.4311	85.4311
35	0.455654	41.5677	33.3525	soil 2	5.3	23	24.6264	33.8403	67.2367	0	67.2367	83.4455	83.4455
36	0.455654	40.8397	35.0481	soil 2	5.3	23	23.7735	32.6683	64.4757	0	64.4757	81.1518	81.1518
37	0.417071	36.5918	36.7047	soil 1	0	38	33.3673	45.8517	58.6875	0	58.6875	83.5631	83.5631
38	0.417071	35.6829	38.323	soil 1	0	38	31.6196	43.4501	55.6135	0	55.6135	80.6058	80.6058
39	0.417071	34.6164	39.9783	soil 1	0	38	29.7759	40.9165	52.3708	0	52.3708	77.3365	77.3365
40	0.417071	33.3806	41.6747	soil 1	0	38	27.8367	38.2518	48.9601	0	48.9601	73.7397	73.7397
41	0.417071	31.9621	43.4172	soil 1	0	38	25.8025	35.4565	45.3823	0	45.3823	69.7972	69.7972
42	0.417071	30.3448	45.2114	soil 1	0	38	23.6737	32.5312	41.6381	0	41.6381	65.4871	65.4871
43	0.417071	28.5094	47.0643	soil 1	0	38	21.4507	29.4765	37.7282	0	37.7282	60.7831	60.7831
44	0.417071	26.4326	48.984	soil 1	0	38	19.1342	26.2932	33.6538	0	33.6538	55.6528	55.6528
45	0.417071	24.0857	50.981	soil 1	0	38	16.725	22.9827	29.4165	0	29.4165	50.0562	50.0562
46	0.417071	21.4325	53.0682	soil 1	0	38	14.2249	19.5472	25.0192	0	25.0192	43.9432	43.9432
47	0.417071	18.4264	55.2622	soil 1	0	38	11.6364	15.9902	20.4665	0	20.4665	37.248	37.248
48	0.417071	14.8597	57.5854	soil 1	0	38	8.87705	12.1984	15.6131	0	15.6131	29.5933	29.5933
49	0.417071	9.53346	60.0686	soil 1	0	38	5.3461	7.34634	9.40288	0	9.40288	18.6882	18.6882
50	0.417071	3.29389	62.7565	soil 1	0	38	1.69359	2.32724	2.97872	0	2.97872	6.26793	6.26793

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.4044

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	29.9322	25	0	0	0
2	30.3674	24.8995	3.05631	0	0
3	30.8027	24.81	7.3278	0	0
4	31.238	24.7311	12.6346	0	0
5	31.6732	24.6627	18.799	0	0
6	32.1085	24.6049	25.6559	0	0
7	32.5437	24.5574	33.0518	0	0
8	32.979	24.5201	40.8434	0	0
9	33.4143	24.4931	48.897	0	0
10	33.8495	24.4762	57.0878	0	0
11	34.2848	24.4695	65.2989	0	0
12	34.7201	24.4729	73.4212	0	0
13	35.1553	24.4864	81.3524	0	0
14	35.5906	24.5101	88.9968	0	0
15	36.0258	24.544	96.2651	0	0
16	36.4611	24.5881	103.074	0	0
17	36.8964	24.6425	109.344	0	0
18	37.3316	24.7074	115.005	0	0
19	37.7669	24.7828	119.988	0	0
20	38.2022	24.8688	124.23	0	0
21	38.6374	24.9657	127.675	0	0
22	39.0727	25.0735	130.27	0	0
23	39.5079	25.1925	131.967	0	0
24	39.9432	25.323	132.722	0	0
25	40.3785	25.465	132.5	0	0
26	40.8137	25.619	131.265	0	0
27	41.249	25.7853	128.992	0	0
28	41.6843	25.9641	125.658	0	0
29	42.1195	26.156	121.247	0	0
30	42.5548	26.3613	115.749	0	0
31	42.9622	26.5661	110.589	0	0
32	43.3695	26.7836	104.48	0	0
33	43.7769	27.0142	97.4257	0	0
34	44.1843	27.2586	89.441	0	0
35	44.5917	27.5174	80.5462	0	0
36	44.999	27.7914	70.7707	0	0
37	45.4064	28.0813	60.1534	0	0
38	45.819	28.3921	55.0017	0	0
39	46.2316	28.7214	48.9535	0	0
40	46.6442	29.0705	42.059	0	0
41	47.0568	29.4408	34.3826	0	0
42	47.4694	29.834	26.0056	0	0
43	47.882	30.2525	17.0298	0	0
44	48.2947	30.6985	7.58205	0	0
45	48.7073	31.1753	-2.1792	0	0
46	49.1199	31.6866	-12.0554	0	0
47	49.5325	32.2373	-21.7956	0	0
48	49.9451	32.8337	-31.0784	0	0
49	50.3577	33.4842	-39.1452	0	0
50	50.7703	34.2005	-44.7508	0	0
51	51.1829	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.37415

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	29.9431	25	0	0	0
2	30.3594	24.9061	3.7191	1.47536	21.6381
3	30.7758	24.8221	8.76481	3.47698	21.6381
4	31.1922	24.748	14.91	5.91476	21.6381
5	31.6086	24.6836	21.9407	8.70382	21.6381
6	32.0249	24.6287	29.6638	11.7676	21.6382
7	32.4413	24.5834	37.9047	15.0367	21.6381
8	32.8577	24.5475	46.505	18.4484	21.6381
9	33.274	24.521	55.3213	21.9458	21.6381
10	33.6904	24.5039	64.2232	25.4772	21.6381
11	34.1068	24.4961	73.0923	28.9956	21.6381
12	34.5232	24.4976	81.8212	32.4583	21.6381
13	34.9395	24.5084	90.3122	35.8266	21.6381
14	35.3559	24.5285	98.4766	39.0654	21.6381
15	35.7723	24.558	106.234	42.1428	21.6381
16	36.1886	24.5969	113.512	45.03	21.6381
17	36.605	24.6453	120.245	47.7009	21.6381
18	37.0214	24.7032	126.374	50.1322	21.6381
19	37.4378	24.7708	131.846	52.3031	21.6381
20	37.8541	24.848	136.616	54.1951	21.6381
21	38.2705	24.9352	140.64	55.7917	21.6382
22	38.6869	25.0323	143.885	57.079	21.6382
23	39.1032	25.1396	146.32	58.0447	21.6381
24	39.5196	25.2573	147.918	58.6787	21.6381
25	39.936	25.3855	148.66	58.9731	21.6381
26	40.3524	25.5245	148.53	58.9214	21.6381
27	40.7687	25.6746	147.517	58.5195	21.6381
28	41.1851	25.8359	145.615	57.7651	21.6381
29	41.6015	26.009	142.824	56.6579	21.6381
30	42.0178	26.194	139.147	55.1994	21.6382
31	42.4342	26.3914	134.595	53.3936	21.6381
32	42.8899	26.6223	129.54	51.3883	21.6381
33	43.3455	26.8691	123.456	48.9748	21.6381
34	43.8012	27.1328	116.377	46.1664	21.6381
35	44.2568	27.4141	108.345	42.9801	21.6381
36	44.7125	27.714	99.4147	39.4376	21.6381
37	45.1681	28.0336	89.6523	35.5649	21.6381
38	45.5852	28.3446	85.338	33.8534	21.6381
39	46.0023	28.6742	80.2082	31.8184	21.6381
40	46.4194	29.0239	74.328	29.4857	21.6381
41	46.8364	29.3952	67.7746	26.886	21.6381
42	47.2535	29.7898	60.6393	24.0555	21.6381
43	47.6706	30.21	53.0302	21.0369	21.6381
44	48.0876	30.6582	45.0754	17.8813	21.6381
45	48.5047	31.1378	36.9278	14.6492	21.6381
46	48.9218	31.6524	28.7713	11.4135	21.6381
47	49.3388	32.2073	20.8295	8.263	21.6381
48	49.7559	32.8088	13.3783	5.30715	21.6381
49	50.173	33.4656	6.82994	2.70942	21.6381
50	50.5901	34.19	2.25098	0.892958	21.6381
51	51.0071	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	20, 20 70, 20 70, 24 70, 31 70, 35 50, 35 30, 25 20, 25
Material Boundary	40, 27 52, 24 70, 24
Material Boundary	30, 25 40, 27 50, 29 54, 31 70, 31